

ASSIGNMENT NO. 1: Introduction to Databases and SQL Environment

Q1) Write SQL commands to create a database named SchoolDB. The program should:
1. Create a new database named SchoolDB. 2. Select the database for use. 3. Display confirmation by creating a simple table inside it.

ANS:

CREATE DATABASE SchoolDB;

USE SchoolDB;

CREATE TABLE TestTable (ID INT);

*Q2) Create a table named Student with the following columns: Column Name Data Type
StudentID INT Constraint StudentName PRIMARY KEY VARCHAR(50) NOT NULL Age
ClassName VARCHAR(30) INT VARCHAR(20) City The program should: 1. Create the
table. 2. Define proper data types. 3. Apply primary key on StudentID.*

ANS:

CREATE TABLE Student (

StudentID INT PRIMARY KEY,

StudentName VARCHAR(50) NOT NULL,

Age INT,

ClassName VARCHAR(20),

City VARCHAR(30)

);

*Q3) Insert at least 10 records into the Student table. The records should include: 1.
Student ID 2. Student name 3. Age 4. Class name 5. City After inserting data, display all
records using the SELECT command.*

ANS:

INSERT INTO Student (StudentID, StudentName, Age, ClassName, City)

VALUES

(1, 'Aarav Shah', 18, 'FYBsc IT' , 'Mumbai'),

(2, 'Meera Joshi', 19, 'FYBCA' , 'Pune'),

(3, 'Rohan Patil', 20, 'SYBsc IT' , 'Nashik'),

(4, 'Sneha Desai', 18, 'FYBCA' , 'Mumbai'),

(5, 'Amit Kumar', 21, 'TYBsc IT' , 'Delhi'),
 (6, 'Pooja Singh', 19, 'SYBCA' , 'Pune'),
 (7, 'Vikram Rao', 20, 'SYBsc IT' , 'Bengaluru'),
 (8, 'Neha Sharma', 18, 'FYBCA' , 'Nashik'),
 (9, 'Rahul Verma', 21, 'TYBsc IT' , 'Mumbai'),
 (10, 'Kavita Iyer', 19, 'SYBCA' , 'Chennai');

SELECT * FROM Student;

StudentID	StudentName	Age	ClassName	City
1	Aarav Shah	18	FYBsc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBsc IT	Nashik
4	Sneha Desai	18	FYBCA	Mumbai
5	Amit Kumar	21	TYBsc IT	Delhi
6	Pooja Singh	19	SYBCA	Pune
7	Vikram Rao	20	SYBsc IT	Bengaluru
8	Neha Sharma	18	FYBCA	Nashik
9	Rahul Verma	21	TYBsc IT	Mumbai
10	Kavita Iyer	19	SYBCA	Chennai

Q4) Create a table named Course with the following fields:

Column Name Data Type CourseID CourseName INT VARCHAR(50) Duration Fees VARCHAR(20) DECIMAL(10,2)

The program should: 1. Create the table. 2. Add a primary key on CourseID. 3. Insert at least 5 course records. 4. Display all course records.

ANS:

CREATE TABLE Course (

CourseID INT PRIMARY KEY,

CourseName VARCHAR(50),

Duration VARCHAR(20),

FEES DECIMAL(10,2)

);

INSERT INTO Course (CourseID, CourseName, Duration, Fees) VALUES

(1 , 'SQL Mastery' , '2 Months' , 4500.00),

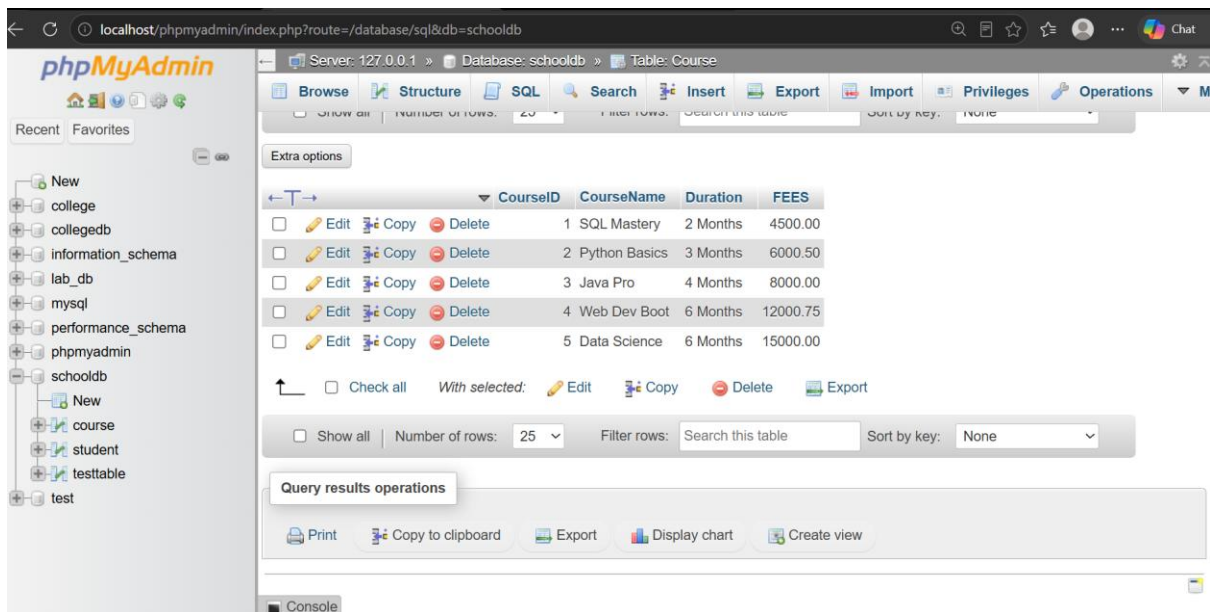
(2 , 'Python Basics' , '3 Months' , 6000.50),

(3 , 'Java Pro' , '4 Months' , 8000.00),

(4 , 'Web Dev Boot' , '6 Months' , 12000.75),

(5 , 'Data Science' , '6 Months' , 15000.00);

SELECT * FROM Course;



The screenshot shows the phpMyAdmin interface for a database named 'schooldb'. The 'Course' table is selected, and its data is displayed in a table format. The table has four columns: CourseID, CourseName, Duration, and FEES. There are five rows of data, each with a unique CourseID and corresponding CourseName, Duration, and FEES.

CourseID	CourseName	Duration	FEES
1	SQL Mastery	2 Months	4500.00
2	Python Basics	3 Months	6000.50
3	Java Pro	4 Months	8000.00
4	Web Dev Boot	6 Months	12000.75
5	Data Science	6 Months	15000.00

Q5) Write SQL queries to display only selected columns from the Student table. Display: 1. StudentName 2. ClassName 3. City Expected output should show only these three columns.

ANS:

SELECT StudentName, ClassName, City

FROM Student;

The screenshot shows the phpMyAdmin interface with the 'Student' table selected in the 'schooldb' database. The table structure is as follows:

Field	Type	Null	Key	Default	Extra
StudentID	int(11)	NO	PRI	NULL	
StudentName	varchar(50)	NO		NULL	
Age	int(11)	YES		NULL	
ClassName	varchar(20)	YES		NULL	
City	varchar(30)	YES		NULL	

The data in the table is as follows:

StudentName	ClassName	City
Aarav Shah	FYBSc IT	Mumbai
Meera Joshi	FYBCA	Pune
Rohan Patil	SYBSc IT	Nashik
Sneha Desai	FYBCA	Mumbai
Amit Kumar	TYBSc IT	Delhi
Pooja Singh	SYBCA	Pune
Vikram Rao	SYBSc IT	Bengaluru
Neha Sharma	FYBCA	Nashik
Rahul Verma	TYBSc IT	Mumbai
Kavita Iyer	SYBCA	Chennai

Q6) Write SQL commands to view the structure of the Student table. Use the suitable command according to the database environment: Database Command MySQL DESC Student; PostgreSQL tudent SQLite PRAGMA table_info(Student);

ANS:

DESC Student; #MYSQL Environment

\d Student #PostgreSQL Environment

PRAGMA table_info(Student) #SQLite Environment

The screenshot shows the phpMyAdmin interface with the SQL query 'DESC Student;' executed successfully. The query results are as follows:

Field	Type	Null	Key	Default	Extra
StudentID	int(11)	NO	PRI	NULL	
StudentName	varchar(50)	NO		NULL	
Age	int(11)	YES		NULL	
ClassName	varchar(20)	YES		NULL	
City	varchar(30)	YES		NULL	

Q7) Prepare a CSV file named students.csv and import it into the Student table. The CSV file should contain: StudentID StudentName 1 Aarav Shah Age ClassName City 18 2 Meera Joshi FYBSc IT Mumbai 19 3 Rohan Patil 20 FYBCA Pune SYBSc IT Nashik The task should include: 1. Creating a matching table. 2. Importing the CSV file. 3. Displaying imported records using SELECT.

ANS:

LOAD DATA INFILE "D:/SEM 3/Students.csv"

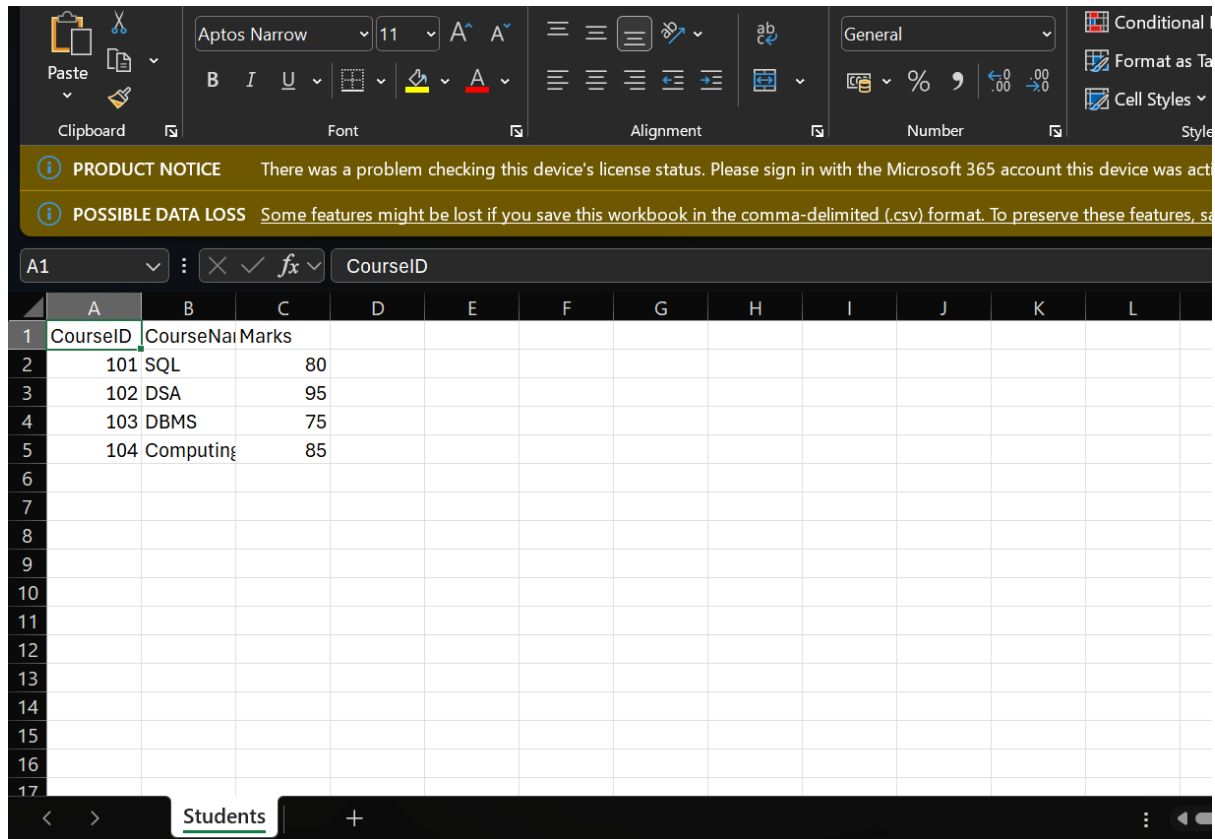
INTO TABLE Student

FIELDS TERMINATED BY ','

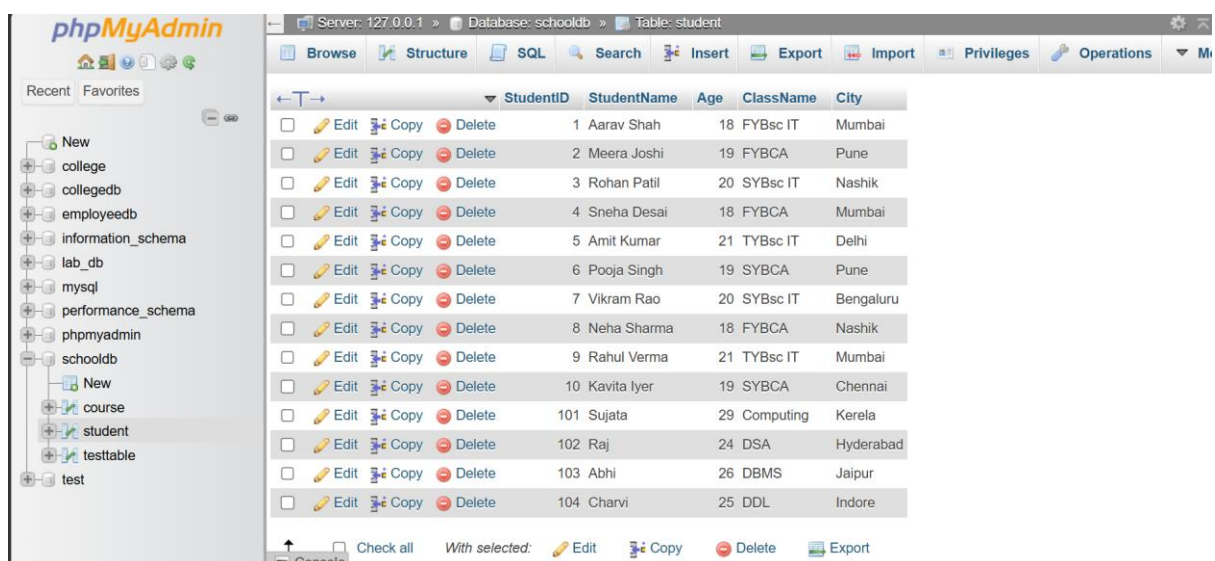
ENCLOSED BY '"'

LINES TERMINATED BY '\n'

IGNORE 1 ROWS;



	A	B	C	D	E	F	G	H	I	J	K	L
1	CourseID	CourseName	Marks									
2	101	SQL	80									
3	102	DSA	95									
4	103	DBMS	75									
5	104	Computing	85									
6												
7												
8												
9												
10												
11												
12												
13												
14												
15												
16												
17												



StudentID	StudentName	Age	ClassName	City
1	Aarav Shah	18	FYBsc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBsc IT	Nashik
4	Sneha Desai	18	FYBCA	Mumbai
5	Amit Kumar	21	TYBsc IT	Delhi
6	Pooja Singh	19	SYBCA	Pune
7	Vikram Rao	20	SYBsc IT	Bengaluru
8	Neha Sharma	18	FYBCA	Nashik
9	Rahul Verma	21	TYBsc IT	Mumbai
10	Kavita Iyer	19	SYBCA	Chennai
101	Sujata	29	Computing	Kerala
102	Raj	24	DSA	Hyderabad
103	Abhi	26	DBMS	Jaipur
104	Charvi	25	DDL	Indore

Q8) Create a database named EmployeeDB and a table named Employee. The table should contain: Column Name Data Type EmployeeID INT EmployeeName VARCHAR(50) Department VARCHAR(30) Salary DECIMAL(10,2)

ANS:

CREATE DATABASE EmployeeDB;

USE EmployeeDB;

CREATE TABLE Employee (

EmployeeID INT PRIMARY KEY, EmployeeName VARCHAR(50),

Department VARCHAR(30), Salary DECIMAL(18,2), City VARCHAR(30)

);

INSERT INTO Employee VALUES

(101, 'John Doe', 'IT', 75000.00, 'New York'),

(102, 'Jane Smith', 'HR', 65000.00, 'Chicago'),

(103, 'Sam Brown', 'Finance', 82000.00, 'Boston'),

(104, 'Lisa Ray', 'IT', 78000.00, 'Seattle'),

(105, 'Mark Lee', 'Sales', 60000.00, 'Austin'),

(106, 'Anna Kim', 'Marketing', 68000.00, 'Denver'),

(107, 'Paul Jay', 'Finance', 81880.00, 'Miami'),

(108, 'Emma Watson', 'HR', 66000.00, 'Dallas');

SELECT * FROM Employee;

The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a database tree. The main area displays a table with the following data:

EmployeeID	EmployeeName	Department	Salary	City
101	John Doe	IT	75000.00	New York
102	Jane Smith	HR	65000.00	Chicago
103	Sam Brown	Finance	82000.00	Boston
104	Lisa Ray	IT	78000.00	Seattle
105	Mark Lee	Sales	60000.00	Austin
106	Anna Kim	Marketing	68000.00	Denver
107	Paul Jay	Finance	81880.00	Miami
108	Emma Watson	HR	66000.00	Dallas

Q9) Write SQL queries to compare the difference between `SELECT *` and selecting specific columns. The task should include: 1. Query using `SELECT * FROM Student`; 2. Query using `SELECT StudentName, City FROM Student`; 3. Write two lines explaining the difference between both outputs.

ANS:

--query 1: the wildcard approach

Select * FROM Student

--query2: the specific approach

Select StudentName, City

From student;

Difference in Outputs

- **SELECT *** returns all columns from the Student table (e.g., ID, Name, Age, City, Phone, etc.).
- **SELECT StudentName, City** returns only the specified columns, making the result more focused and efficient.

In practice, using specific columns is preferred because it reduces unnecessary data retrieval and improves query performance.

StudentID	StudentName	Age	ClassName	City
1	Aarav Shah	18	FYBsc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBsc IT	Nashik
4	Sneha Desai	18	FYBCA	Mumbai
5	Amit Kumar	21	TYBsc IT	Delhi
6	Pooja Singh	19	SYBCA	Pune
7	Vikram Rao	20	SYBsc IT	Bengaluru
8	Neha Sharma	18	FYBCA	Nashik
9	Rahul Verma	21	TYBsc IT	Mumbai
10	Kavita Iyer	19	SYBCA	Chennai
101	Sujata	29	Computing	Kerela
102	Raj	24	DSA	Hyderabad
103	Abhi	26	DBMS	Jaipur
104	Charvi	25	DDL	Indore

Q10) Create a complete SQL script for a mini college database. The script should include: 1. Creating database MiniCollegeDB. 2. Creating tables Student, Course, and Faculty. 3. Adding suitable columns and data types. 4. Inserting at least 5 records in each table. 5. Displaying all records from each table. 6. Viewing the structure of each table.

ANS:

CREATE DATABASE MiniCollegeDB;

USE MiniCollegeDB;

CREATE TABLE Student (SID INT PRIMARY KEY, SName VARCHAR(50));

CREATE TABLE Course (CID INT PRIMARY KEY, CName VARCHAR(50));

CREATE TABLE Faculty (FID INT PRIMARY KEY, FName VARCHAR(50));

INSERT INTO Student

VALUES (1,'Ali'), (2, 'Ben'), (3, 'Cam'), (4, 'Dan'), (5, 'Eve');

INSERT INTO Course

VALUES (10, 'Math'), (11, 'Sci'), (12, 'Art'), (13, 'Lit'), (14, 'CS');

INSERT INTO Faculty

VALUES (100, 'Dr.X'),(101, 'Dr.Y'),(102,'Dr.Z'), (103, 'Dr.A'), (104, 'Dr.B');

SELECT * FROM Student; SELECT FROM Course; SELECT FROM Faculty;

DESC Student; DESC Course; DESC Faculty;

The screenshot shows the phpMyAdmin interface for a database named 'schooldb'. The 'Faculty' table is selected, and its structure and data are displayed. The table has two columns: 'FID' (Faculty ID) and 'FName' (Faculty Name). The data is as follows:

FID	FName
100	Dr.X
101	Dr.Y
102	Dr.Z
103	Dr.A
104	Dr.B

The interface includes a sidebar with a tree view of databases and tables. The main panel shows the 'Browse' tab with a SQL query editor containing 'SELECT * FROM `Faculty`'. Below the query editor, there are options for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (Search this table), and 'Sort by key' (None). The 'Extra options' section shows the table structure and data. At the bottom, there are checkboxes for 'Check all', 'With selected', and buttons for 'Edit', 'Copy', 'Delete', and 'Export'.